

JOSE OSA

(980) 254-3443 | joseosa@umich.edu | Ann Arbor, MI

EDUCATION

University of Michigan, Ann Arbor, MI

- Aerospace and Aeronautics Engineering Graduate student

Columbia University, New York City, NY

- Bachelor of Science in Materials Science Engineering and Aerospace Engineering Minor, Conferred May 2026. GPA: 3.55, Dean's List Award.

The University of the South, Sewanee, TN

- Bachelor of Arts/Science Physics and Mathematics Minor, Conferred May 2024. GPA: 3.87. Honors/Awards: Global Honors Award
- President of the Sigma Pi Sigma Physics Society and Member of the Order of the Gown.

Space Architecture Studio, Jeffrey Montes, Online

- Space Architecture Diploma, Conferred Fall 2025. Semester studio course taught by Blue Origin's Space Architect Jeffrey Montes, where I architected a Dual Person Spacecraft and presented to industry professionals.

TECHNICAL SKILLS

Aerospace, Propulsion, Spacecraft Technology, Aerodynamics, Rocketry, Systems Engineering, Additive Manufacturing, SolidWorks, Fusion 360, Ansys, GMAT, STK, Nanofabrication, MATLAB, 2PP Printing, NanoScribe, Laser Machining, Laser Nucleation, Circuit Design, Metallurgy, Crystallography, Grain Structure, Ultra Thin Metal Films, Ceramics and Polymer Composites, Photoresins, Mechanical Properties Testing, SEM, Python, Data analysis, Molecular Modeling, Gamess, Burnman Code, Mathematica, LaTeX, Excel, Arduino, Welding and soldering, Adobe Photoshop, Illustrator, Premiere Pro.

PROFESSIONAL EXPERIENCE

3D Printed Micro-Rocket Combustion Engineering Researcher, Aubin Group, Internship, Zoetic Robotics Lab, Fall 2026-Present, Ann Arbor, MI.

- Aerospace engineering research intern with the modeling, printing, and testing of small scale resin-based 3D-printed combustion and rocket systems.

Mineral Physics and Engineering Intern, Wentzcovitch Group, Internship, Lamont-Doherty Earth Observatory, Summer 2025, Palisades, NY.

- Engineering researcher for the Dr. R. Wentzcovitch Minerals project group at the Lamont-Doherty Earth Observatory in collaboration with Columbia University's Geodynamics and Seismology department.

Research Assistant, Kysar Group, Internship, Columbia University, Spring 2025-Fall 2025, New York, NY.

- Research assistant for the nanofabrication and testing of medical devices at the nano and micro scale. Researcher at the Advanced Science Research Center (ASRC) in the City College of New York.

CSA Peer Tutor, Part-time, Columbia University, Academic year 2024-2025, New York, NY

- Peer tutor for undergraduate students in Mathematics Calculus Courses

Engineering Research Assistant, Internship, dolaGon, Fall 2022-March 2023, Remote

- Main research assistant in the vehicle engineering project for the Terrain and Trial Locations team.

Science Coordinator and counselor, Internship, Skylemar, Summer 2022, Naples, MN

- Science coordinator for the entire camp activities.

ACADEMIA EXPERIENCE

L3Harris Guardian Project

- Development of a GEO and HEO satellite architecture for countering and safeguarding potential spying operations by foreign satellite systems.

Columbia Space Initiative

- **Blue Skies:** NASA Engineering competition group from CSI, which aims to tackle agriculture problems via aeronautical vehicles.
- **RASC-AL:** NASA Engineering competition group from CSI, which aims to design from scratch an entire self-sufficient habitat Mars mission for consideration by future NASA operations in Mars.

Nucleation Growth Stage Characterization of Ultra-Thin Metal Films via Laser Shocking and AFM Imaging for Aerospace Materials

- Laser-induced nucleation and growth phenomena in ultra-thin metallic films of Ni, Al, Cu, using controlled pulsed laser exposure and high-resolution atomic force microscopy (AFM). The resulting framework connects thin-film microstructural evolution to broader aerospace-relevant phenomena

Argon Plasma Physics Seminar

- Argon Plasma analysis, Plasma element identification, Plasma data acquisition and analysis

3D Harmonic Oscillator in Quantum Mechanics with Applications to Group 6 Hexafluorides

- Experimented with computational analysis using Mathematica to model three dimensional quantum atoms. Pending Paper publishing.

ENGINEERING PORTFOLIO

<https://joseosa.github.io/Engineering-Portfolio/index.html>

LANGUAGES

Fluent in English | Fluent in Spanish | German | French